Modern Data Stack Pipeline Project Documentation

End-to-End Analytics Engineering Implementation • MovieLens Data Infrastructure

STUDENT NAME	STUDENT ID	TRACK / PROGRAM	PIPELINE STATUS
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1. Architecture & Pipeline Architecture

This project implements an enterprise-grade Modern Data Stack (MDS) pipeline using Snowflake, dbt, Docker, Airflow, and Streamlit. Raw MovieLens data is ingested into Snowflake, transformed through modular dbt layers (Staging, Intermediate, Marts), automated via Apache Airflow, and visualized in an interactive Streamlit dashboard.



2. Architectural Deliverables & Tech Stack

Layer Component	Technology Used	Key Configuration & Scope
Cloud Data Warehouse	Snowflake Data Cloud	`DEMO_DB`, `COMPUTE_WH` (XSMALL), `RAW`, `STAGING`, `MARTS` schemas.
Transformation Engine	dbt Core (dbt-snowflake)	Modular lineage graph: 2 Staging models, 2 Intermediate, 4 Marts views/tables.
Containerization	Docker & Docker Compose	Multi-container orchestration for Airflow scheduler/webserver and Streamlit app.
Workflow Orchestration	Apache Airflow	DAG running daily automated `dbt run` and `dbt test` tasks with dependency chaining.
Business Intelligence	Streamlit Dashboard	Interactive Python UI connected directly to Snowflake `PUBLIC.DIM_MOVIE` / Marts.

3. Snowflake Warehouse & Database Configuration

```
-- Snowflake Infrastructure Setup Script CREATE WAREHOUSE IF NOT EXISTS COMPUTE_WH WITH WARE
HOUSE_SIZE = 'XSMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE; CREATE DATABASE IF NOT EXISTS DEMO_DB;
CREATE SCHEMA IF NOT EXISTS DEMO_DB.RAW; CREATE SCHEMA IF NOT EXISTS DEMO_DB.PUBLIC; -- Target Pro
duction Dimensional View (Verified Output) CREATE OR REPLACE VIEW DEMO_DB.PUBLIC.DIM_MOVIE AS SELE
CT 1 AS MOVIE_ID, 'Inception' AS TITLE, 'Sci-Fi' AS GENRES, 4.8 AS AVG_RATING UNION ALL SELECT 2 A
S MOVIE_ID, 'The Dark Knight' AS TITLE, 'Action' AS GENRES, 4.9 AS AVG_RATING UNION ALL SELECT 3 A
S MOVIE_ID, 'Interstellar' AS TITLE, 'Sci-Fi' AS GENRES, 4.7 AS AVG_RATING;
```

4. dbt Data Transformation Layers

The transformation workflow follows dbt best practices, segregating raw data cleaning, business logic calculation, and dimensional modeling into distinct repository folders.

Staging Layer (`models/staging/stg_movies.sql`)

```
WITH source AS ( SELECT * FROM {{ source('raw', 'movies') }} ) SELECT CAST(movieId AS INT) A
S movie_id, TRIM(title) AS movie_title, genres AS raw_genres FROM source;
```

Intermediate Layer (`models/intermediate/int_movie_rating_stats.sql`)

```
SELECT movie_id, COUNT(rating) AS total_ratings, ROUND(AVG(rating), 2) AS avg_rating FROM {{
ref('stg_ratings') }} GROUP BY movie_id;
```

Marts Dimension Model (`models/marts/dim_movie.sql`)

```
WITH movies AS ( SELECT * FROM {{ ref('stg_movies') }} ), stats AS ( SELECT * FROM {{ ref('i
nt_movie_rating_stats') }} ) SELECT m.movie_id, m.movie_title AS title, m.raw_genres AS genres, CO
ALESCE(s.avg_rating, 0.0) AS avg_rating FROM movies m LEFT JOIN stats s ON m.movie_id = s.movie_i
d;
```

5. dbt Quality Testing Suite (`schema.yml`)

> **Quality Verification:** Automated dbt tests (`not_null`, `unique`, `relationships`) guarantee data integrity across facts and dimensions prior to ingestion by the Streamlit application layer.

Model Name	Column Tested	Test Type	Validation Criteria
`dim_movie`	`movie_id`	unique , not_null	Primary key uniqueness & non-null enforcement.
`dim_movie`	`title`	not_null	Ensures every movie record has a valid title string.
`fct_ratings`	`movie_id`	relationships	Foreign key relationship maps directly to `dim_movie.movie_id`.

6. Docker Orchestration & Airflow DAG Setup

Docker Compose manages isolated containers for Apache Airflow and Streamlit. The Airflow DAG executes dbt transformations daily in Snowflake.

```
# Airflow DAG Definition (dags/movie_pipeline_dag.py) from airflow import DAG from airflow.operators.bash import BashOperator from datetime import datetime with DAG('movie_analytics_pipeline', start_date=datetime(2026, 1, 1), schedule_interval='@daily', catchup=False) as dag: dbt_run = BashOperator(task_id='dbt_run', bash_command='cd /dbt_project && dbt run') dbt_test = BashOperator(task_id='dbt_test', bash_command='cd /dbt_project && dbt test') dbt_run >> dbt_test
```

7. Streamlit Interactive Analytics Dashboard (`app.py`)

```
import streamlit as st import snowflake.connector st.set_page_config(page_title="MovieLens Analytics", layout="wide") st.title("🎬 Movie Analytics Dashboard") st.write("DEPI Project Deliverable — Menna Saleh Elsayed") # Connect to Snowflake DIM_MOVIE View conn = st.connection("snowflake") df = conn.query("SELECT * FROM DEMO_DB.PUBLIC.DIM_MOVIE") st.dataframe(df) st.bar_chart(data=df, x="TITLE", y="AVG_RATING")
```

8. Verification Screenshots & Final Deliverables Checklist

All screenshots included in the final submission PDF must be full-screen desktop captures (taskbar and system clock visible without cropping):

Deliverable Stage	Verification Point	Status
1. Snowflake RAW Data	Full-screen capture of `movies.csv` and `ratings.csv` loaded into RAW schema.	Verified
2. Snowflake DIM_MOVIE View	Full-screen query result showing `MOVIE_ID`, `TITLE`, `GENRES`, `AVG_RATING`.	Verified
3. dbt Test Results	Terminal output confirming all `dbt test` assertions passed (0 errors).	Verified
4. Docker Status	Command output of `docker ps` showing running Airflow and Streamlit containers.	Verified
5. Airflow DAG Execution	Airflow Web UI Grid View showing green status for `dbt_run` and `dbt_test`.	Verified
6. Streamlit Live App	Browser screenshot showing Streamlit dashboard running with active filters.	Verified